

MIDTERM REPORT

GENERATIVE AI IN VISUAL DESIGN: *Investigating the Role of Human Involvement in Visual Communication*

1. Project Overview

My senior seminar project explores how different levels of human involvement affect the quality and effectiveness of AI-supported visual design. As generative AI becomes more commonly used in creative work, I am interested in understanding what the human designer still contributes to the process and whether that contribution can be seen or measured in the final design.

The project compares three design conditions: **human-only, AI-assisted, and AI-only**. Each condition responds to the same three promotional flyer briefs, creating a total of nine flyers.

The project focuses on three main areas: **visual quality**, including hierarchy, contrast, alignment, spacing, readability, and overall visual appeal; **communication effectiveness**, including how easily viewers can locate, understand, and remember important information; and **efficiency and value**, including production time, iterations, and the role of human decision-making during the design process.

The expected final outcome is a research paper comparing the three conditions using participant evaluations, task-based testing, and information collected during the design process. The goal is not only to determine which condition performs better, but also to better understand **what human involvement contributes when AI becomes part of the design process**.

2. Work Completed: Weeks 1–7

Project Development and Initial Research

During the first few weeks of the semester, I worked on turning my general interest in AI and design into a research question that could realistically be studied within one semester.

One of the first decisions I made was to avoid treating the project as simply “human versus AI.” That comparison leaves out an important middle ground because AI is increasingly being incorporated into human creative workflows. Because of this, I developed three conditions: human-only, AI-assisted, and AI-only.

I also conducted background research on visual design principles and ways of evaluating design quality. This helped me identify some of the factors I want to measure, including visual hierarchy, readability, organization, clarity, spacing, contrast, and visual appeal.

Another important decision was separating **visual quality from communication effectiveness**. A flyer can be visually attractive while still failing to communicate important information clearly. Because of this, the eventual participant study will not only ask participants which designs they prefer. It will also test how successfully they can find and understand information within each flyer.

Developing the Experimental Method

A large portion of my progress involved defining what each condition actually means.

For the **human-only condition**, the designer completes the flyer without using generative AI as part of the creative process.

For the **AI-assisted condition**, AI can be used during the design process, but the human designer is still responsible for making decisions and producing the final design. AI use and important human changes will be documented so that I can better understand how the collaboration occurred.

For the **AI-only condition**, the AI receives the same underlying brief and generates the flyer without a human independently constructing the design.

Originally, I considered describing human involvement using percentages, such as saying a design was a certain percentage human and a certain percentage AI. I realized that this would be difficult to measure consistently. Instead, I decided to focus on **observable human decisions**, such as changing layouts, modifying hierarchy, selecting typography, rejecting AI suggestions, adjusting content, or refining an AI-generated result.

I also developed the basic plan for participant testing. Participants will view the designs without being told which condition produced them. The designs will be presented in a way that reduces potential ordering and condition bias. I plan to use ratings for areas such as readability, organization, clarity, professionalism, and visual appeal, along with tasks or questions that test whether participants can successfully locate and recall information.

After data collection, the results will be organized and statistically analyzed to determine whether meaningful differences exist among the three conditions.

Creating the Standardized Design Briefs

I developed three standardized design briefs so that every condition is solving the same underlying communication problems.

The three scenarios are:

1. **Culture Fest**: a campus cultural event.
2. **Know Your Numbers**: a health-information campaign.
3. **FutureTech Fellowship**: a career and recruitment opportunity.

Each brief contains the scenario, target audience, required information, and communication objective. However, the briefs do not tell the designer what colors, typography, imagery, layout, or visual style to use.

This was intentional. If I provided too much visual direction, I would already be making many of the design decisions myself. Keeping the factual requirements consistent while leaving the visual solution open allows each condition to respond to the same communication problem differently.

I also created the fictional institution **Riverton University** so that existing university branding would not influence the designs.

Habeebah Dawodu 1	Habeebah Dawodu 2
VISUAL DESIGN RESEARCH BRIEFS 1. Brief 1 - Event Promotion: Culture Fest Scenario Riverton University is hosting Culture Fest, an evening celebrating the cultures represented across campus. Target Audience Undergraduate and graduate students at Riverton University. Required Information - Event name: Culture Fest - Date: Friday, October 23, 2026 - Time: 6:00-9:00 PM - Location: University Quad - Admission: Free - Activities include international food tasting, live performances, cultural showcases, music, and student organization booths. - The first 100 attendees will receive a free Culture Fest tote bag. - Students are encouraged to wear clothing representing their culture. - Registration is encouraged but not required. - A QR code/link will be provided for registration. Communication Objective Encourage students to attend Culture Fest and communicate the essential information they need to participate. 2. Brief 2 - Health/Public Awareness: Know Your Numbers Scenario Riverton University Student Health Services is running a campus awareness campaign encouraging students to understand the importance of knowing their blood pressure and to receive a free blood pressure screening. Target Audience Undergraduate and graduate students at Riverton University. Required Information - Campaign name: Know Your Numbers - High blood pressure often has no noticeable symptoms. - Knowing your blood pressure can help identify potential health concerns early.	- Riverton University students can receive a free blood pressure screening. - Location: Student Wellness Center - Dates: October 26-30, 2026 - Hours: 10:00 AM-4:00 PM - Screening takes approximately 5 minutes. - No appointment is required. - Students should bring their university ID. - Contact: wellness@riverton.edu Communication Objective Help students understand why knowing their blood pressure matters and encourage them to receive a free screening. 3. Brief 3 - Career/Recruitment: FutureTech Fellowship Scenario Riverton University's Career Development Center is promoting FutureTech Fellowship, a paid summer fellowship that gives students professional experience working with technology companies. Target Audience Riverton University undergraduate students interested in technology, regardless of major. Required Information - Program name: FutureTech Fellowship 10-week paid summer fellowship - Fellows work on projects with participating technology companies. - Program dates: June 7-August 13, 2027 - Open to Riverton University sophomores, juniors, and seniors of any major. - No previous professional/technology experience is required. - Fellows receive a \$6,000 stipend. - Application requires a résumé and a short statement of interest. - Application deadline: February 12, 2027 20 fellows will be selected. - An application link/QR code will be provided. - Questions: careers@riverton.edu Communication Objective Make eligible students aware of the fellowship, help them understand the opportunity and eligibility requirements, and encourage interested students to apply.

Figure 1. Example of one of the standardized design briefs provided across the experimental conditions.

Completing the AI-Only Flyers

The **AI-only condition is currently complete**, with all three AI flyers generated from the standardized briefs.

For these flyers, I provided the AI with the information contained in each brief and avoided adding unnecessary aesthetic instructions. This was important because if I provided extremely detailed instructions about color, hierarchy, typography, layout, or visual style, I would effectively be making many of the design decisions myself through the prompt.

The purpose of this condition is instead to create a baseline representing what the AI produces when it receives the communication problem and required information.

At this point, **three of the planned nine experimental flyers are complete**.



Figure 2. AI-only output for the Culture Fest design brief.



Figure 3. AI-only output for the Know Your Numbers design brief.



Figure 4. AI-only output for the FutureTech Fellowship design brief.

Creating these flyers also made me think more critically about **prompting**. Even when I intentionally avoid providing visual direction, the way that a brief is communicated to an AI can still influence what it produces. Prompting has therefore become an important issue that I plan to return to when interpreting the eventual results.

Recruiting Designers and Beginning the Remaining Conditions

By Week 7, I had recruited **three designers** to participate in the remaining design conditions.

Currently, one designer is working under the human-only condition, one has selected the AI-assisted condition, and the third is still determining which condition they will complete. All three have received information about the project and are beginning their work.

I also developed separate instructions for the human-only and AI-assisted conditions so that each designer understands what is and is not allowed.

For the human-only condition, the designer cannot use generative AI during the creative process. For the AI-assisted condition, AI can be used as part of the workflow, but the designer must remain involved in the process and make the final design decisions.

Habeebah Dawodu 3

Human-Only Designer Instructions**Your Task:**

Create a promotional flyer based on the design brief provided. You may determine the wording, information hierarchy, layout, typography, colors, imagery, and overall visual presentation.

Design Requirements:

- Keep all required factual information from the brief accurate.
- You may use the design platform of your choice (Canva, Figma, Adobe Express, etc.).
- You may use the platform's normal/manual design tools, including typography, shapes, images, and other standard design elements.
- **Do not use generative AI at any stage of the design process.** This includes AI-generated layouts, images, copy, design suggestions, or other generative features built into your chosen platform.
- The final design decisions should be entirely your own.
- Please record the **platform used** and your approximate **total design time**.

AI-Assisted Designer Instructions**Your Task:**

Create a promotional flyer based on the design brief provided. You may determine the wording, information hierarchy, layout, typography, colors, imagery, and overall visual presentation.

Design Requirements:

- Keep all required factual information from the brief accurate.
- You may use the design platform of your choice (Canva, Figma, Adobe Express, etc.).
- **You may use generative AI as part of your design process.**
- AI may be used for things such as brainstorming, copy suggestions, layout ideas, visual generation, or other parts of the creative process.
- You must remain responsible for selecting, modifying, arranging, and/or approving the final design.
- Please document:
 - Platform used
 - AI tool(s) used
 - What you used AI for
 - Prompts used, where applicable
 - Major changes you made to AI outputs
 - Approximate total design time

Figure 5. Example of the instructions developed to distinguish the human-only and AI-assisted experimental conditions.

Change in Design Platform Requirement

One methodological change I made during Weeks 1–7 concerns the software used by the human designers.

My original plan was to have designers work in **Figma** so that the design platform would remain consistent across the experiment. However, after beginning designer recruitment, I found that some participants were more comfortable using other platforms, particularly **Canva**.

I decided to allow designers to use the design platform they are most comfortable with rather than requiring everyone to use Figma. This change reduces the possibility that a designer's unfamiliarity with a specific program will affect their performance. If a designer normally works in Canva but is required to use Figma for the experiment, the resulting differences could reflect software familiarity rather than the effect of human or AI involvement.

The design platform used by each designer will therefore be **recorded as part of the process data**.

This change also requires an additional control for the human-only condition because platforms such as Canva now include built-in generative AI features. Human-only designers may use the standard manual tools available within their chosen platform but cannot use generative AI features. AI-assisted designers may use AI features, but the way those features are used will be documented.

This represents a change from controlling the **software environment** to controlling the **type of AI involvement**, which is more directly related to the main research question.

Continued Research and Ideas for the Final Paper

As the experiment has developed, my background research and conversations about the project have raised additional questions that I want to consider later in the final paper.

One involves the **restrictions and prior instructions that already exist within an AI system**. AI does not necessarily approach a design problem from a blank slate. Its training, system-level instructions, capabilities, and restrictions can influence what it produces before a user begins making design decisions.

This makes me interested in whether some patterns that appear in AI-generated design are characteristics of AI design in general or characteristics of the particular system and constraints being used.

Another area I want to explore is **prompting as part of the design process**. Prompting is usually treated as simply the instruction given to the AI, but writing, changing, and refining a prompt can itself involve creative decision-making. If changing a prompt significantly changes a design, prompting may represent another form of human involvement that needs to be considered.

I am also interested in the possibility of an AI being given a more specialized **design skill or designer-specific context**. For example, an AI that has access to a designer's previous work, visual preferences, or established design system may produce something very different from a general-purpose AI responding to a brief for the first time.

These are not findings from my experiment. They are questions that have emerged while developing the project and may help me interpret my eventual results, discuss limitations, and identify directions for future research.

3. Evidence of Progress

My progress through Week 7 is supported by several concrete project artifacts. These include the three standardized design briefs, the three completed AI-only flyers, instructions developed for the human-only and AI-assisted conditions, documentation of the experimental methodology, and recruitment of three designers who are currently beginning the remaining designs.

My current project status is:

Project Component	Midterm Status
Project topic and research questions	Completed
Three-condition methodology	Completed
Evaluation approach	Completed
Culture Fest brief	Completed
Know Your Numbers brief	Completed
FutureTech Fellowship brief	Completed

AI-only flyers	3 of 3 completed
Designer recruitment	3 designers recruited
Human-only flyers	In progress
AI-assisted flyers	In progress
Participant testing	Not started
Data analysis	Not started
Final research paper	In progress throughout semester

The figures included throughout this report provide visual evidence of the briefs, completed AI outputs, and experimental instructions rather than relying only on descriptions of the work completed.

4. Progress Compared with Original Proposal

Compared with my original proposal, most of the foundational work for the experiment has been completed. I have finalized the research direction, defined the three experimental conditions, developed the three design briefs, planned the evaluation method, completed the AI-only condition, and recruited designers for the remaining conditions.

The main difference between my original timeline and my current progress is that I expected to be further into production and participant testing by this point.

Recruiting designers and making sure that the three conditions were clearly defined took more time than expected. I also realized that I needed to be more careful about how I defined AI assistance, prompting, and human involvement before moving into data collection.

Another change from the original proposal is allowing designers to choose their preferred design platform rather than requiring Figma. I made this change because differences in software familiarity could affect design performance and interfere with the variable I am actually interested in: **the level and type of AI involvement**.

Because the human-only and AI-assisted flyers are still being created, participant testing has not started yet. I would rather complete the experimental materials correctly than begin testing with an incomplete or inconsistent set of designs.

The project is therefore roughly aligned with the original testing timeline and the experimental structure is more clearly defined than it was at the beginning of the semester.

5. Challenges and Solutions

One of my biggest challenges has been defining **AI-assisted design**. There are many different ways a designer can use AI, from generating an entire layout to using it for ideas, copy, visual elements, or revisions. This made it difficult to define AI assistance using a simple percentage.

I addressed this by focusing instead on the actual decisions made during the process. The AI-assisted condition allows AI use, but the human designer remains responsible for choosing, editing, rejecting, and finalizing the design. This challenge has largely been resolved through clearer condition instructions, although the actual AI interactions will still need to be documented.

Another challenge was keeping the three conditions comparable without making every flyer look the same. I addressed this by standardizing the **design problem instead of the design solution**. Every condition receives the same required information and communication objective, but the visual interpretation remains open.

Designer recruitment also took longer than expected. Since the project depends on real human design processes, I needed to find designers willing to participate and follow the experimental rules. Three designers have now been recruited, so this issue is currently being resolved as production moves forward.

The design platform also became a methodological challenge. Requiring Figma would standardize the software, but it could disadvantage designers who are more experienced in another platform. I addressed this by allowing platform choice and documenting which software each designer uses. This introduces platform choice as a potential limitation, but it reduces the effect of software unfamiliarity on designer performance.

Finally, creating the AI-only flyers revealed that **prompting can influence the result even when the goal is to reduce human involvement**. I addressed this in the current experiment by limiting unnecessary aesthetic direction. Prompt influence cannot be completely removed, so it will also be acknowledged as a limitation and discussion point in the final paper.

6. Current Project Status

At the midpoint of the semester, the project has moved from planning into active experimental production.

The experimental structure is established, the three conditions have been defined, all three standardized briefs are complete, the evaluation approach has been planned, and all three AI-only flyers have been completed.

Three designers have also been recruited, and work on the human-only and AI-assisted conditions is currently underway. Designers are now permitted to work in their preferred design platforms, with their platform choice and AI usage documented as part of the experiment.

The major portions that remain are completing the remaining flyers, conducting participant testing, organizing and analyzing the resulting data, and writing the results, discussion, limitations, and conclusion of the final paper.

Although I have not reached participant testing yet, I now have the framework and materials needed to move into that stage once the remaining designs are completed.

7. Plan for the Second Half of the Semester

My first priority after the midterm is to complete the remaining human-only and AI-assisted designs so that I have the full set of **nine flyers**.

For each designer, I will document the platform used, production time, and relevant information about the design process. For AI-assisted work, I will also document how AI was incorporated into the workflow.

Once all of the flyers are complete, I will check them for factual consistency and prepare them for participant testing. The designs will be presented without identifying their condition so that participants' responses are not influenced by knowing whether a human or AI created them.

I will then conduct the user study and collect ratings related to visual quality and communication effectiveness, along with task-based responses that measure how successfully participants can find or remember important information.

After data collection, I will organize the results and conduct the appropriate statistical analysis to compare the three conditions.

The final paper will move beyond simply identifying which condition receives the highest scores. I want to use the results to think more carefully about **why** the conditions may differ and what those differences say about human involvement.

In particular, I plan to return to the questions that have emerged during the first half of the project: the effect of AI's built-in constraints, prompting as a possible design activity, the decisions humans make when collaborating with AI, the influence of the design platform, and how a general-purpose AI might differ from an AI that has learned or been given a particular designer's visual language.

Ultimately, I want the final project to address a larger question that has become increasingly important as I have worked on the study: **when we describe a design as AI-assisted, what exactly belongs to the AI, what belongs to the human designer, and how does that relationship affect the final communication?**

Works Cited

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